# TITAN : A DECENTRALIZED THREAT INTELLIGENCE SHARING USING BLOCKCHAIN

## A PROJECT REPORT

submitted by

**Malavika R MUT23CC043**

to

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**Department of Cyber Security**
**Muthoot Institute of Technology and Science**
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# DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

**Malavika R - MUT23CC043**

Place : Varikoli

Date : 27th March 2026

# CERTIFICATE

*This is to certify that the report entitled* **"TITAN : A DECENTRALIZED THREAT INTELLIGENCE SHARING USING BLOCKCHAIN"**, *submitted by* **Malavika R** *to Muthoot Institute of Technology and Science, Varikoli for the award of the degree of Bachelor of Technology in Computer Science & Engineering(Cyber Security) is a bonafide record of the project work carried out by them, under our supervision and guidance. The content of the report, in full or parts have not been submitted to any other Institute or University for the award of any other degree or diploma.*

Dr. Haritha H
Asst. Professor
Dept. of Cyber Security
Project Guide

Dr. Vicky Nair
Head of the Department

Ms. Sreeja S , Ms. Sneha Thankachan
Asst. Professor,
Dept. of Cyber Security
Project Coordinators

Place : Varikoli
Date : 27th March 2026

# ACKNOWLEDGMENT

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# ABSTRACT

Cyberattacks are rapidly increasing, making it essential for organizations to share Cyber Threat Intelligence (CTI) to improve overall security. However, most existing CTI sharing systems are centralized and depend heavily on manual trust between organizations. This often results in the spread of fake or low-quality threat data and creates hesitation among organizations to share sensitive information. To address these challenges, the proposed system introduces a decentralized, blockchain-based CTI sharing model. Blockchain technology ensures transparency, immutability, and tamper-evident records for all shared intelligence. In addition, a reputation scoring mechanism is used to evaluate the reliability of participants and encourage honest contributions. This approach improves trust, data quality, and collaboration among organizations, enabling a more secure and efficient cyber threat intelligence sharing environment

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# LIST OF FIGURES

# CHAPTER 1

# INTRODUCTION

## 1.1 INTRODUCTION

Cyberattacks are increasing rapidly, making Cyber Threat Intelligence (CTI) sharing very important. However, existing CTI systems are centralized, which leads to trust issues, data manipulation, and single point failures. The lack of transparency and proper verification also reduces confidence among organizations. As a result, effective collaboration in sharing threat information becomes difficult.

TITAN introduces a decentralized, blockchain-based CTI sharing system with a reputation scoring mechanism. Blockchain ensures secure, transparent, and tamper-proof data records. The reputation system helps identify trustworthy contributors and filter low-quality data. This approach improves trust, security, and overall efficiency in cyber threat intelligence sharing.

## 1.2 SCOPE AND MOTIVATION

The increase in cyberattacks highlights the need for secure CTI sharing. Existing centralized systems face trust issues and lack transparency. This motivates the use of blockchain for a reliable solution. The project develops a decentralized CTI sharing platform. It ensures secure and transparent data exchange. A reputation system identifies trusted contributors, improving collaboration and security.

# CHAPTER 2

# PROPOSED WORK

## 2.1 OBJECTIVES

- Build a decentralized threat intelligence sharing system.
- Implement an automated EigenTrust model to objectively score and verify user reputation.
- Ensure secure and transparent data with smart contracts.
- Improve efficiency and encourage quality contributions.
- Reduce cost and delay with efficient blockchain use.

## 2.2 PROBLEM STATEMENT

Centralized Cyber Threat Intelligence (CTI) platforms face several critical challenges. They are vulnerable to single points of failure, making them easy targets for DDoS attacks and data manipulation. Additionally, the lack of trust due to manual verification processes makes the system slow and prone to collusion or Sybil attacks by malicious contributors. Furthermore, high transaction costs and latency reduce efficiency, preventing the real-time sharing of critical threat information.

## 2.3 LITERATURE SURVEY

### 2.3.1 Towards Improved Trust in Threat Intelligence Sharing Using Blockchain and Trusted Computing

Wu et al. proposed a blockchain-based CTI sharing system with trusted computing to improve trust and transparency. The system uses reputation scoring to verify data reliability. It reduces data tampering and enhances auditability. However, it has limitations like scalability issues and high complexity. The study mainly focuses on trust models rather than real-world implementation. It provides a strong base for decentralized CTI systems. Overall, it highlights the importance of blockchain in secure data sharing.

### 2.3.2 Blockchain-based Secure Information Sharing Framework for Cyber Threat Intelligence

Li et al. developed a blockchain-based secure CTI sharing framework using smart contracts and off-chain storage. The system ensures tamper-proof data and removes the need for a central authority. It improves data integrity and security. However, it suffers from high transaction cost and delay in blockchain operations. Scalability is also a challenge in large networks. This work supports the use of Ethereum in CTI sharing. It emphasizes secure and decentralized architecture.

### 2.3.3 A Reputation-based Trust Management Model for Information Sharing.

Zhang and Liu proposed a reputation-based trust management model for information sharing. It uses trust scores and peer evaluation to identify reliable users. The system promotes honest participation and detects malicious contributors. However, it is vulnerable to collusion attacks. It also requires continuous monitoring. The model improves trust but has some limitations in security. It forms the basis for reputation systems in CTI.

### 2.3.4 Decentralized Cyber Threat Intelligence Sharing Using Smart Contracts.

Patel and Shah introduced a decentralized CTI sharing system using blockchain and smart contracts. It removes single point of failure and ensures transparency. The system encourages users to contribute and prevents freeriding. It improves overall trust in the network. However, it faces issues like scalability and blockchain latency.

Energy consumption is also a concern. This work directly relates to decentralized CTI platforms.

### 2.3.5 EigenTrust Algorithm for Reputation Management in Distributed Systems.

Kamvar et al. proposed the EigenTrust algorithm for reputation management in distributed systems. It calculates global trust scores using an eigenvector-based approach. The system collects local trust values from different users and combines them to generate a global trust ranking. It effectively identifies malicious nodes and reduces the impact of dishonest participants in the network. This improves the reliability and security of peer-to-peer systems. The algorithm also helps in maintaining consistency in trust evaluation. However, it is computationally complex and requires multiple iterations for accurate results. It is also sensitive to initial trust values. Despite these limitations, it is widely used for trust calculation in decentralized environments.

### 2.3.6 Blockchain Technology for Secure Data Sharing in Cybersecurity.

Dorri et al. studied the use of blockchain technology for secure data sharing in cybersecurity. The system uses distributed ledger architecture along with cryptographic hashing and consensus mechanisms to ensure data security. It provides immutability, transparency, and removes the need for a central authority. This helps prevent data tampering and improves trust among participants. The approach can be applied to various cybersecurity and data-sharing systems. However, it has challenges such as high energy consumption and scalability issues. Transaction speed may decrease as the network grows. It also requires significant computational resources. Overall, it provides a strong theoretical and practical foundation for blockchain-based CTI sharing systems.

## 2.4 EXISTING SYSTEM AND PROPOSED SOLUTION

### 2.4.1 EXISTING SYSTEM

In the current Cyber Threat Intelligence (CTI) sharing environment, information exchange is mainly handled through centralized platforms. A central server is responsible for collecting, storing, and distributing threat intelligence among participating organizations. While this approach provides a simple and structured system, it introduces several limitations. One major issue is the presence of a single point of failure, where any attack such as a Distributed Denial of Service (DDoS) can disrupt the entire system. Additionally, centralized systems are more vulnerable to data manipulation, unauthorized access, and potential data breaches.

Another significant drawback of the existing system is the lack of trust among participants. Trust is usually built manually through prior interactions or agreements, which is slow and not always reliable. There is no proper automated mechanism to verify the authenticity or quality of the shared threat data. As a result, the system may contain fake, outdated, or low-quality information. Some participants may also engage in freeriding, where they consume information without contributing any valuable data. This reduces the overall effectiveness of the system.

Furthermore, the lack of transparency and auditability makes it difficult for organizations to track and verify the origin of shared intelligence. The central authority controls access and validation, which may lead to bias or misuse of power. High latency and inefficiency in data sharing also prevent real-time updates, which are critical in handling fast-evolving cyber threats. These challenges highlight the need for a more secure, transparent, and decentralized approach, which is addressed in the proposed TITAN system.

| **Feature** | **Existing System** | **Proposed TITAN System** |
|---|---|---|
| Architecture | Centralized | Decentralized (Blockchain-based) |
| Single Point of Failure | Yes | No |
| Data Security | Vulnerable to attacks | Tamper-proof and secure |
| Trust Mechanism | Manual and unreliable | Automated reputation-based trust |
| Transparency | Limited | High (Transparent ledger) |
| Auditability | Difficult to track | Easily traceable |
| Real-time Sharing | Limited | Faster and efficient |
| Data Quality | Unverified / low-quality possible | Verified and reliable data |

Table 2.1: Comparison of Existing System and Proposed TITAN System

### 2.4.2 PROPOSED SYSTEM

The proposed system, TITAN, introduces a decentralized framework for Cyber Threat Intelligence (CTI) sharing using blockchain technology. Unlike traditional systems, it eliminates the need for a central authority and allows organizations to share threat data in a distributed and secure manner. Each transaction is recorded on a blockchain, ensuring that the data is transparent, tamper-proof, and easily verifiable by all participants. This approach removes the single point of failure and significantly improves system reliability and availability.

A key feature of the proposed system is the use of smart contracts to automate various processes such as user registration, data submission, and validation. These smart contracts ensure that all operations are executed in a secure and consistent manner without human intervention. Additionally, the system incorporates a reputation scoring mechanism inspired by EigenTrust, which evaluates the trustworthiness of participants based on their contributions and feedback from others. This helps in identifying reliable contributors and filtering out malicious or low-quality data.

The system also uses a combination of on-chain and off-chain storage to improve efficiency. While the blockchain stores hash values for integrity verification, the actual CTI data is stored off-chain to reduce storage cost and improve performance. This design helps in minimizing transaction costs and latency, enabling faster sharing of threat intelligence. Furthermore, the system encourages active participation by rewarding high-quality contributions and discouraging freeriding behavior.

Overall, the proposed TITAN system enhances trust, security, and collaboration among organizations. It provides a transparent and automated mechanism for CTI sharing, ensuring data integrity and reliability. By addressing the limitations of existing centralized systems, it offers a scalable and efficient solution that can be applied to real-world cybersecurity environments.

# CHAPTER 3

# PROJECT DESIGN

## 3.1 SYSTEM ARCHITECTURE

- User Registration Module
- Authentication & Access Control
- CTI Data Submission Module
- Blockchain Network (Ethereum)
- Smart Contracts Management
- Off-chain Storage System
- Data Integrity Verification (Hashing)
- Reputation Scoring System (EigenTrust)
- Trust Evaluation & Rating Module
- Data Retrieval & Sharing Module

## 3.2 MODULES

### 3.2.1 Authentication Registration

This module handles user authentication and registration using MetaMask wallet connection. Each user connects their wallet to join the system securely. A unique identity is created based on the user's Web3 address. The system initializes a starting trust score for every new participant. This process is managed through blockchain to avoid manual control. It ensures only valid users can access the platform. Overall, it provides a secure entry point to the system.

### 3.2.2 Off-Chain Storage Hash Generation

This module manages the storage of CTI data efficiently. The actual threat intelligence data is stored off-chain to reduce blockchain cost. A SHA-256 hash is generated for each data entry. This hash acts as a unique representation of the data. Only the hash is stored on the blockchain for verification. This approach ensures data integrity while saving storage space. It also improves system performance and scalability.

### 3.2.3 Smart Contract Registry

This module records important information on the blockchain using smart contracts. It stores the generated hash and the user's Web3 address permanently. This ensures that all data entries are transparent and tamper-proof. Smart contracts automate the recording process without manual interference. Once stored, the data cannot be altered or deleted. This increases trust among participants. It forms the core of secure data management in the system.

### 3.2.4 Reputation Management (EigenTrust)

This module manages the trust level of each participant in the network. It collects ratings from users on shared threat intelligence. Based on these ratings, the system updates trust scores using a reputation model. It helps identify reliable and honest contributors. Malicious or low-quality contributors get lower scores. The updated trust values are shared across the network. This improves overall data quality and trust in the system.

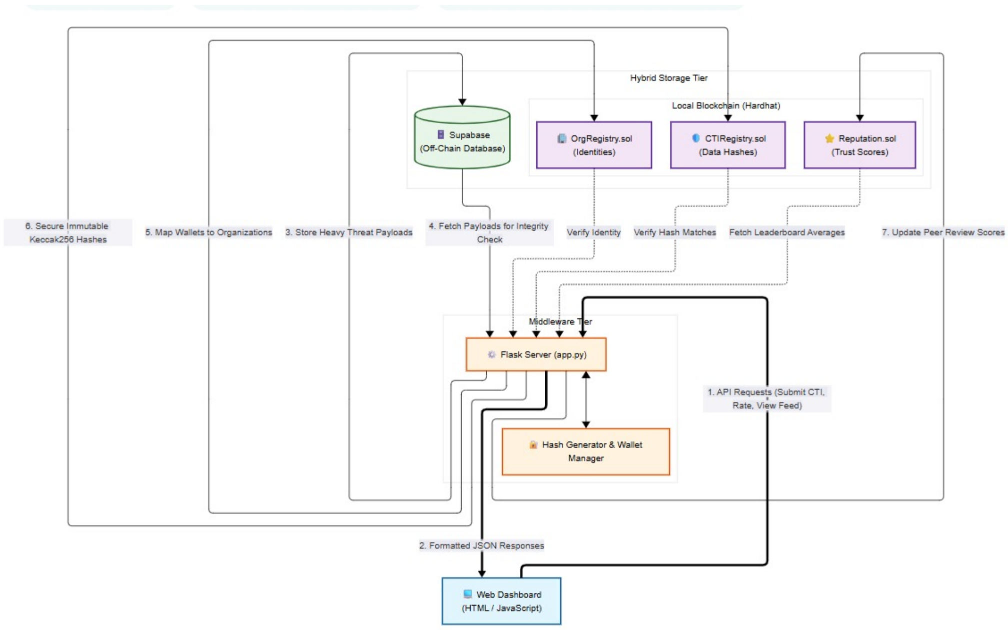


Figure 3.1: System Architecture

## 3.3 DATA FLOW DIAGRAM

### 3.3.1 DFD LEVEL 0

Provides a high-level overview of the entire platform as a single system interacting with external users and organizations.

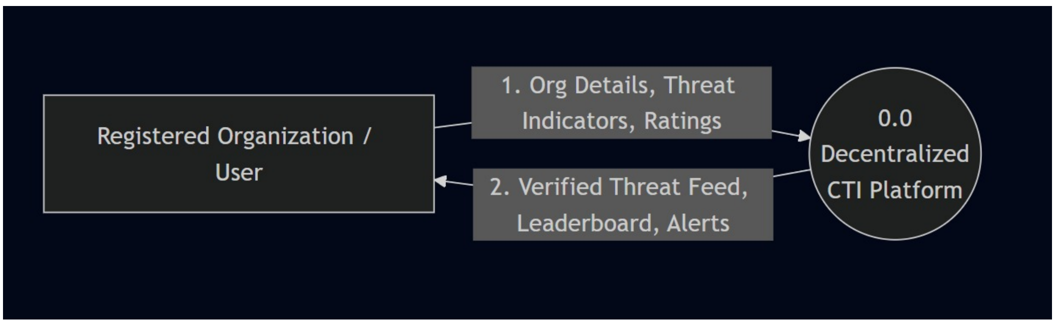


Figure 3.2: DFD Level 0 – Context Diagram

### 3.3.2 DFD LEVEL 1

Breaks down the main system into its core functional modules and maps how data flows between them and your hybrid storage layers.

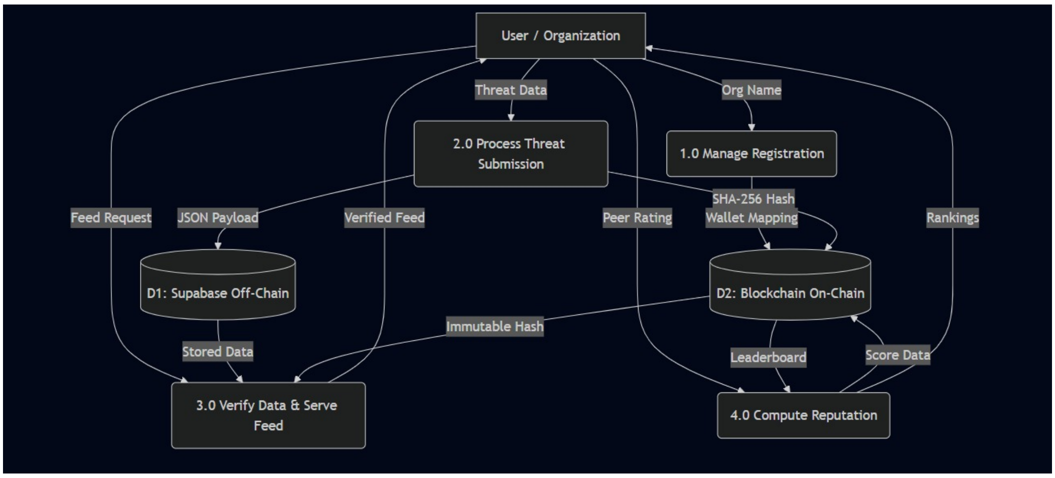


Figure 3.3: DFD Level 1 – System Flow

### 3.3.3 DFD LEVEL 2

Zooms in on one specific module from Level 1 to illustrate its granular, step-by-step programmatic logic and precise data routing.

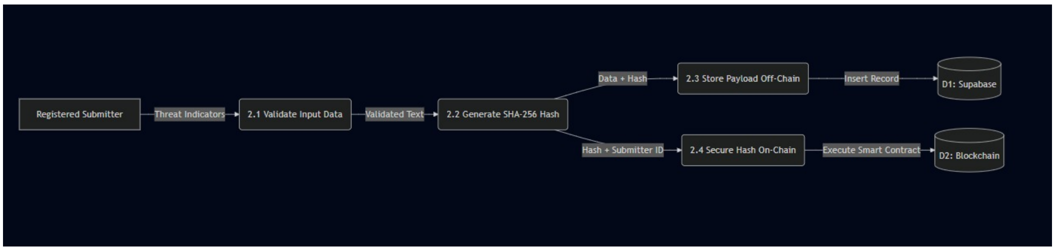


Figure 3.4: DFD Level 2 – Detailed Process Flow

## 3.4 DATABASE TABLE DESIGN

### 3.4.1 CTI Storage Table

Stores threat intelligence details, reporting organization, and integrity hash for secure verification.

### 3.4.2 Organization Table

Stores organization information related to threat reports such as organization name and associated CTI records.

## 3.5 GUI DESIGN

The GUI is divided into sections for organization registration, CTI submission, live feed access, rating, and reputation leaderboard. Each section contains input fields and buttons to allow users to interact with the system and perform specific operations.

## 3.6 TECHNOLOGY STACK

- Smart Contracts & Web3: Solidity, Ethereum, Remix IDE, MetaMask
- Programming Languages: Python, JavaScript
- Operating System: Windows, Linux

## 3.7 SYSTEM REQUIREMENTS

### 3.7.1 Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Recommended 16 GB
- Storage: 256 GB HDD/SSD
- Network: Stable Internet Connection

### 3.7.2 Software Requirements

- Operating System: Windows / Linux
- Blockchain Platform: Ethereum
- Smart Contracts: Solidity
- Programming Languages: Python, JavaScript
- Development Tools: Remix IDE, MetaMask

# CHAPTER 4

# CONCLUSION & FUTURE ENHANCEMENT

## 4.1 CONCLUSION

The TITAN system provides an effective solution to the limitations of traditional Cyber Threat Intelligence (CTI) sharing systems. By using a decentralized approach, it removes the need for a central authority and enables secure collaboration among organizations. It successfully supports intelligence sharing in zero-trust environments without compromising security.

A key advantage of TITAN is the elimination of single points of failure. Unlike centralized systems, it is resistant to attacks such as data breaches and DDoS. Blockchain technology ensures that all shared data is transparent, tamper-proof, and permanently recorded, improving trust and data integrity.

The use of Ethereum smart contracts automates processes like registration, data validation, and access control. This reduces manual effort and increases system efficiency. The EigenTrust-based reputation mechanism helps identify reliable contributors and filters out low-quality or malicious data.

Overall, TITAN offers a secure, transparent, and efficient CTI sharing framework. It improves collaboration, trust, and data quality among organizations. With further improvements, it has strong potential for real-world cybersecurity applications.

## 4.2 FUTURE ENHANCEMENT

- Zero-Knowledge Proofs (ZKPs): Implementing ZKPs to allow organizations to prove they possess valid threat intelligence without revealing their identity.
- Layer 2 Deployment: Migrating the smart contract to a Layer 2 network (like Polygon or Arbitrum) to drastically reduce transaction gas costs.
- AI/ML Integration: Deploying a machine learning model to pre-scan and validate threat submissions automatically before human peers rate them.

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# *APPENDIX*

## CODE SNIPPETS

### Hybrid Data Storage (Off-Chain & On-Chain)

This snippet demonstrates the dual-storage approach, saving the raw threat intelligence to an off-chain database (Supabase) while securing its cryptographic hash on the blockchain.

```
hash_value = w3.keccak(text=cti_text)
cti_hash_hex = hash_value.hex()

1. Store immutable hash on the Blockchain
tx_hash = cti_contract.functions.registerCTI(hash_value).transact({
    "from": sender_wallet
})

2. Store full data payload Off-Chain (Supabase)
db_data = {
    "ctiHash": cti_hash_hex,
    "orgName": org_name_input,
    "payload": {"indicators": cti_text}
}
supabase.table("off_chain_storage").insert(db_data).execute()
```

### Real-Time Data Integrity Verification

This logic recalculates the hash of the off-chain database record and compares it to the immutable hash stored on the blockchain to detect tampering before displaying the feed.

```
for record in cti_records:
    current_data = record["payload"].get("indicators", "")
    stored_hash = record["ctiHash"]

    1. Recalculate the Keccak256 hash of the current database text
    recalculated_hash = w3.keccak(text=current_data).hex()

    2. Compare hashes to detect unauthorized modifications
    if recalculated_hash == stored_hash:
        record["integrity_status"] = "Unchanged"
    else:
        record["integrity_status"] = "Changed"
```

### Secure Peer Rating Validation

This snippet ensures that organizations can only rate threat intelligence provided by their peers, preventing users from artificially inflating their own reputation scores.

```
1. Identify the original author of the threat intelligence
response = supabase.table("off_chain_storage").select("orgName").eq("cti_id", cti_i
original_submitter = response.data[0]["orgName"]

2. Security Check: Prevent Self-Rating
if original_submitter == rater_org_name:
    return jsonify({"error": "Security Violation: You cannot rate your own data."})

3. Submit peer review to the Reputation Smart Contract
tx_hash = rep_contract.functions.submitRating(int(cti_id), int(rating)).transact({
```

```
    "from": rater_wallet
})
```

# SCREENSHOTS

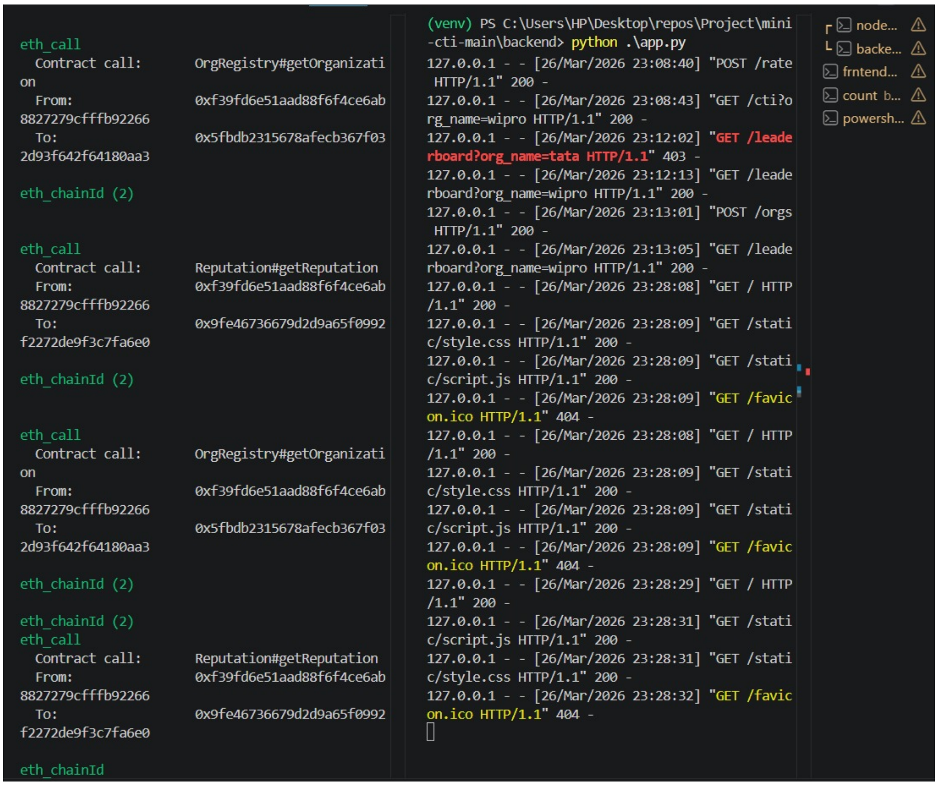

Figure 4.1: The Blockchain and Backend Terminals

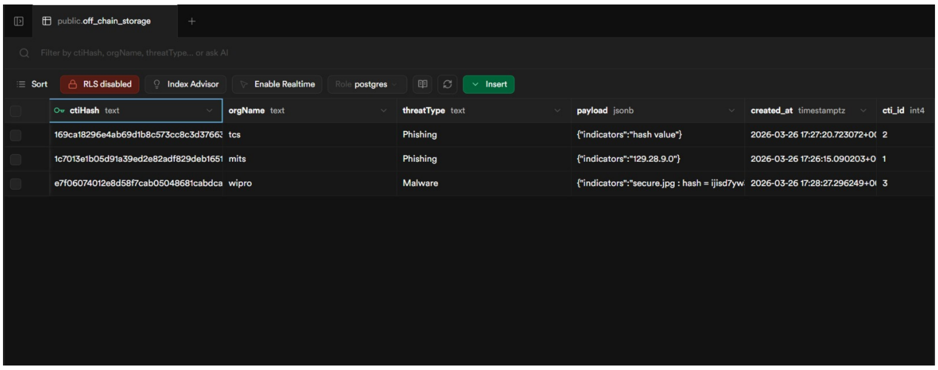

Figure 4.2: The Off-Chain Storage

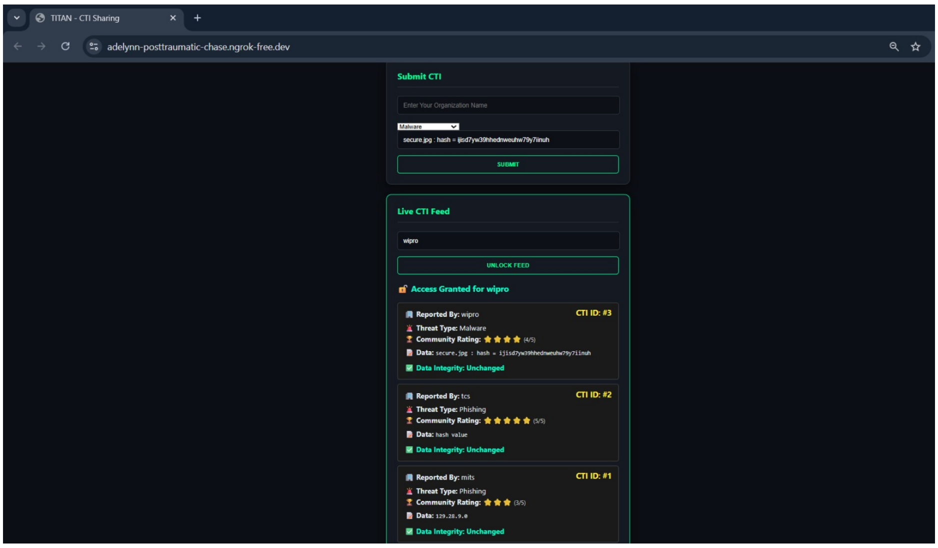


Figure 4.3: The Main Dashboard

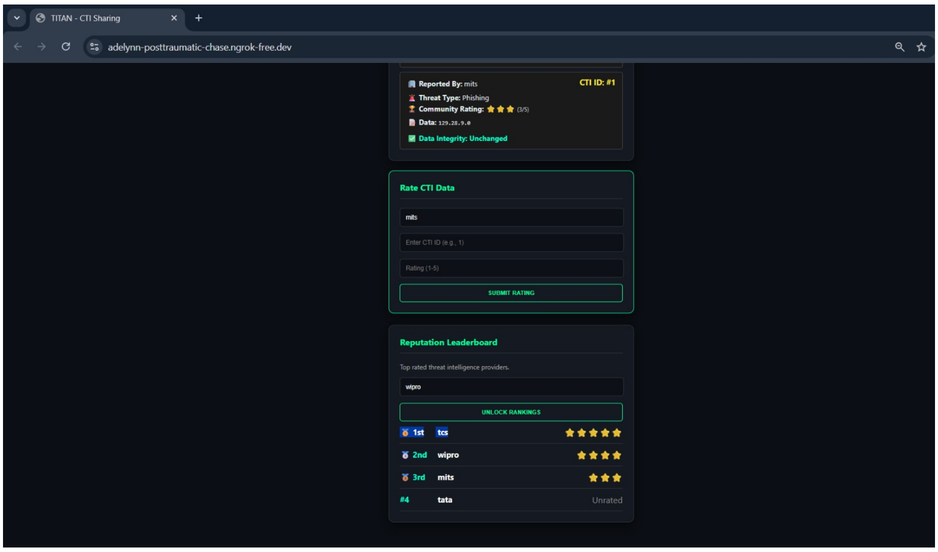


Figure 4.4: The Reputation Leaderboard